

Assignment 1

Data Types & Arrays

Part A: Data Types:

1. Declare the following variables:

- a) A **4-bit signed packed logic vector** named a
- b) An **8-bit unsigned byte** named b
- c) A **16-bit shortint signed** named c
- d) A **bit [3:0]** vector named d
- e) A **real** variable named r
- f) A **string** called myString initialized to "SV Assignment1"

2. Write an initial block to:

- Assign -3 to a and print its decimal value.
- Assign 250 to b and print its value.
- Assign -12345 to c and print it.
- Assign 4'b1101 to d and print its binary after 5ns delay.
- Assign 3.14159 to r and print it after 15ns delay.
- Print the string myString.

Part B: Fixed Arrays: - in the same file

3. Declare and initialize the following fixed arrays:

- a) A **1D unpacked array of int type** with values 1,2,3,4,5
- b) A **packed array of logic type of three elements two bit each**. Assign all elements to 2'b10.
- c) A **1D unpacked array has 4 elements of type byte** with values 10, 20, 30, 40.
- d) Print all the arrays content in separate display statement.

4. Declare a **struct type** called student_t with:

- string name
- int id
- byte grade

Then:

- a) Declare a **fixed array of 3 students of type** student_t called students.
- b) Assign the following:

```
students[0] = {"Ali", 1, 85};
students[1] = {"Sara", 2, 90};
students[2] = {"Omar", 3, 78};
```

c) print each student's details in the format: Name: <name>, ID: <id>, Grade: <grade>

5. **Declare a union type** called data_u with:

- int i;
- byte b;

Then:

- a) Declare a variable of type data_u called data
- b) Assign 32'h12345678 to data.i
- c) Print data.i and data.b as hexadecimal.
- d) Assign 8'h12 to data.i
- e) Print data.i and data.b as hexadecimal.

Part C: Dynamic Arrays: - in the same file

- 1) Declare a dynamic array of type int called dyn_arr.
- 2) Initialize it with 5 elements {5, 10, 15, 20, 25}.
- 3) Declare another dynamic array dyn_copy.
- 4) Copy the contents of dyn_arr into dyn_copy.
- 5) Print both arrays to verify.
- 6) Increase the size of dyn_arr to 20 elements.
- 7) Assign the last 15 new elements with 'hC and keeping its old content.
- 8) Print the updated array.
- 9) Use a for loop to print all elements of dyn_arr.
- 10) Use a foreach loop to print all elements of dyn_copy.
- 11) Delete dyn_copy.
- 12) Print its size after deletion

Deliverables:

U have to submit two separate files

- **One PDF** contains:

- Snippets for the implemented code.
- Snippets for the waveform shows the values for created variables
- Snippets for the logs show the all printed values.

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- **One Zip file** contains:

- Snippets for the implemented code.
- Snippet for the **run.do** file (That you've used for automating the process)

The delivered zip file must be named like **your_name_Assignment_1.rar** for example:

Hassan_Khaled_Assignment_1.rar

Good Luck